FINAL YEAR PROJECT · HKBU COMPUTER SCIENCE

RadAssist AI

An AI-assisted Online System for X-Ray Report Generation

Clinician-in-the-loop chest X-ray reporting inside a hospital CMS-style workspace — AI drafts, the radiologist decides.

PANG Ho Yiu
Student ID 23226609

Demonstration system · not a clinical product

Agenda

1. **Problem & objectives**
   Why a research model is not yet a usable tool
2. **Solution & architecture**
   Roles, components, technology stack
3. **End-to-end workflow**
   Booking → film → AI draft → review → endorse
4. **AI pipeline**
   CURV report generation + Azure OpenAI assistance
5. **Clinical modules**
   Patient chart, Azure findings, Sepsis Monitor, Care Notes
6. **Admin, data & security**
   Users, audit, data model, access control
7. **Testing, limitations, future work**
   How it was verified and what comes next
8. **Live demo & Q&A**

PROBLEM

Background: the deployment gap

Vision-language models such as CURV can now write chest X-ray reports — but they stay in scripts and notebooks, out of reach of the clinicians who need them.

1 Time-consuming reporting

Radiology reports are professional work product. Writing similar reports from a blank page occupies valuable radiologist hours.

Goal: AI drafts automatically when the film is saved; the radiologist reviews instead of typing from scratch.

2 AI is not accessible

Research models have no standard web interface. Running them needs Python, a GPU/Apple Silicon box and a command line.

Goal: browser login, worklist and one review screen — no notebook, no command line.

3 Static, opaque output

Image-in, text-out tools are not interactive. Clinicians cannot see where a finding is, edit it, or tell AI text from human text.

Goal: film + boxes beside an editable report, an AI-vs-manual log while drafting, and a clean signed report.

Project objectives

1. Online prototype

Use the pretrained CURV model to assist radiologists by generating chest X-ray reports.

2. Interactive interface

Film with editable boxes and measurements beside an editable report; Word / PDF export.

3. Complete web information system

Sessions, roles, asynchronous inference, persistence, audit and document generation.

4. Radiologist in the loop

AI output is always a draft until a radiologist endorses it. Doctors read endorsed reports only.

5. Clinical context

Patient Master chart with vitals, labs, care notes and a teaching sepsis screen beside imaging.

6. Run locally, deploy later

Semester 1 on a local Mac (CURV on Apple Silicon); optional hosted deployment in Semester 2.

Solution overview: four roles, one workflow

Role-separated, clinician-in-the-loop web application presented in a hospital CMS-style shell (beige title bar, MODULES column, RIS-style worklist).

Referring doctor

- Books a chest X-ray on a calendar
- Reads endorsed reports only
- Orders labs on the chart
- Exports Word / PDF

Technician

- Sees the requested-case queue
- Registers (uploads) the film
- CURV starts automatically
- Never sees the film or report afterwards

Radiologist

- Works the X-ray worklist
- Edits report, boxes, measurements
- Endorses (signs) the report
- Shares an endorsed report

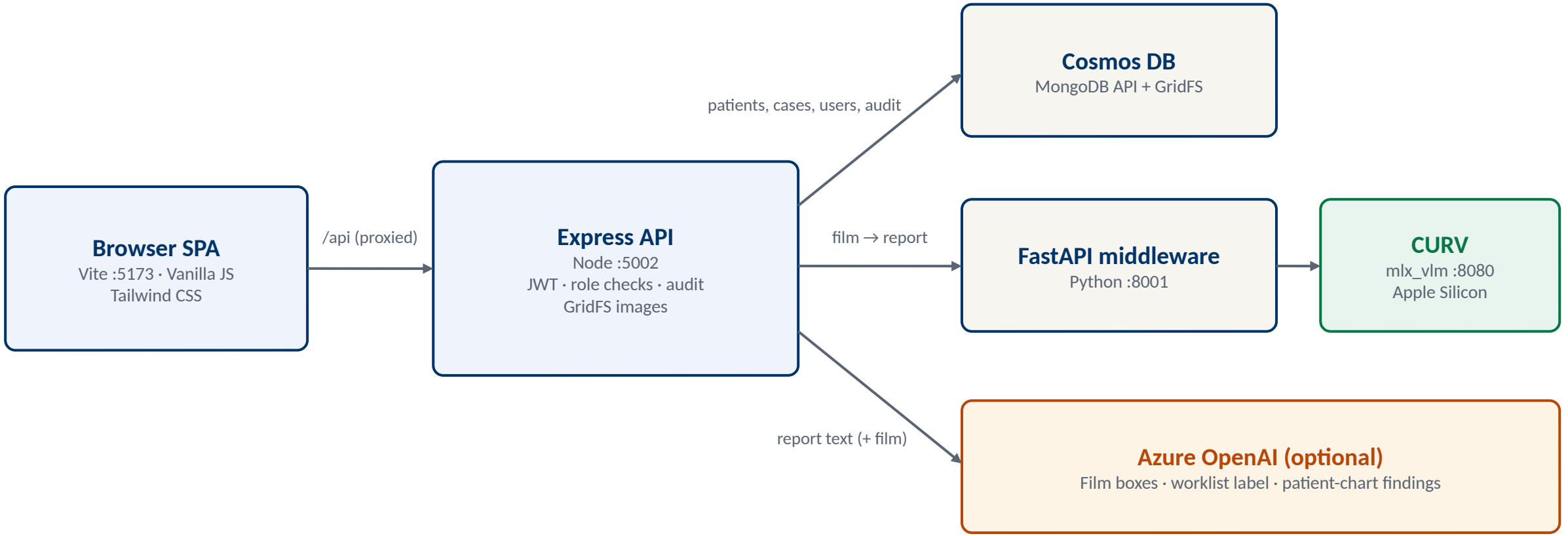
Administrator

- Everything a radiologist can do
- Adds / disables users
- Reads the audit log
- Bulk-deletes test data

Control principle: AI drafts → the radiologist edits and endorses → only then does the referring doctor see the report. AI never signs, and the issued report carries no AI / manual markings.

SOLUTION

System architecture



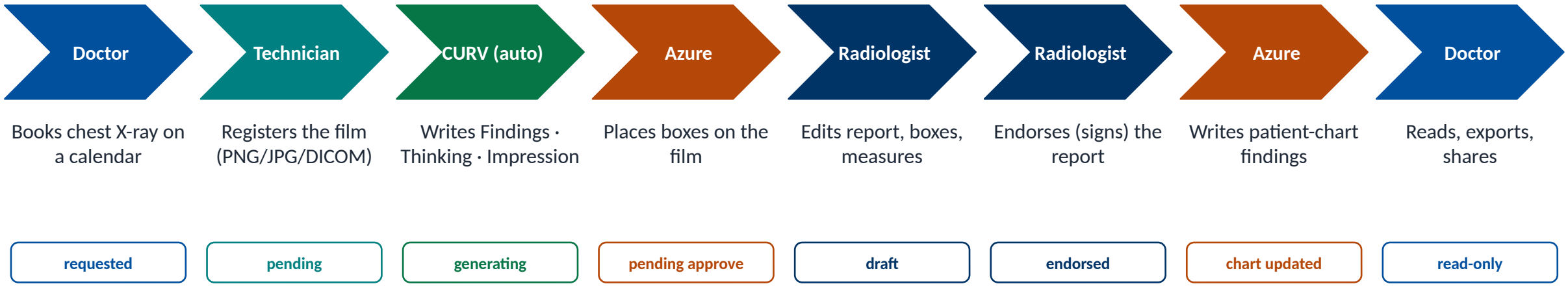
- **Asynchronous AI** — the API stores the film and returns immediately; CURV runs in the background and the worklist moves Generating → Pending approve.
- **Graceful degradation** — if CURV or Azure is down, the case is still saved and the radiologist can write the report by hand.
- **Separation of concerns** — CURV writes the report; Azure only assists (boxes, labels, chart findings) and never edits report text.

SOLUTION

Technology stack

Layer	Technology	Why
Frontend	Vanilla JavaScript SPA, Vite 8, Tailwind CSS 3	Lightweight; tiny DOM helper + pub/sub store; fast dev reload
Backend API	Node.js, Express 4, Mongoose 8, JWT, bcrypt, Helmet	REST gateway, role middleware, audit writer, document streams
Database	Azure Cosmos DB (MongoDB API) + GridFS	Managed, cloud-hosted persistence for charts, cases and films
Report model	CURV (vision-language) served by mlx_vlm on Apple Silicon	Pretrained chest X-ray report generator, runs locally
Model gateway	FastAPI middleware (Python 3.12, uvicorn)	Official CURV prompt, image resizing, DICOM rendering
AI assistance	Azure OpenAI (gpt-4.1-mini, JSON mode)	Film boxes, worklist labels, patient-chart findings
Documents	docx (Word) and browser PDF	Clean endorsed report with film, measurements, signature
Testing	Node test scripts with linkedom (DOM), e2e API script	193 automated UI tests across 19 suites

End-to-end workflow



Automatic first draft

There is no “Use AI” button. Saving the film starts CURV in the background; the API returns at once and the worklist updates itself when the draft is ready.

Human sign-off

Endorse is refused while AI is still running. After endorsement the report, boxes and measurements are locked; remarks and the urgent flag can still change.

Clean issued report

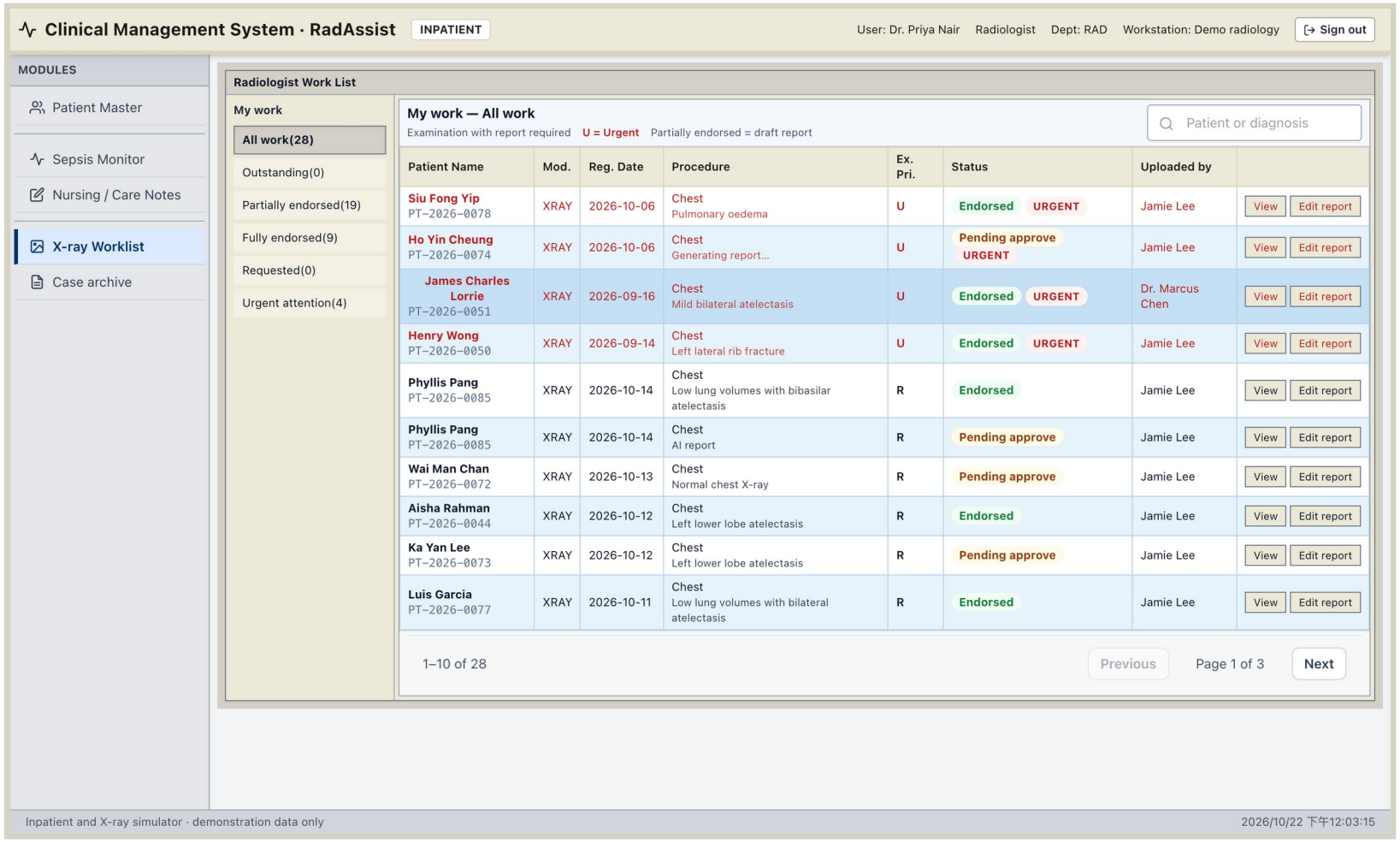
The doctor’s view, Word and PDF contain Findings, Impression, measurements and the signature — no hidden Thinking and no AI / manual tags.

Role-based access control

Action	Technician	Radiologist	Doctor	Admin
Browse Patient Master, Sepsis Monitor, Care Notes	yes	yes	yes	yes
Book (request) a chest X-ray	no	no	yes	yes
Register the film on a request	yes	no	no	yes
See drafts, films and AI boxes	no	yes	no	yes
Edit report, boxes, measurements · re-run AI · endorse	no	yes	no	yes
Read an endorsed report · export Word / PDF	no	yes	yes	yes
See Azure findings on the patient chart	no	yes	yes	yes
Order labs on the chart	no	no	yes	yes
Share an endorsed report (public link)	no	yes	no	yes
Users, audit log, bulk delete	no	no	no	yes

Enforced on the server (route middleware + response stripping), not just hidden in the UI. Technician responses never contain the film, report, findings or chart findings.

Radiologist X-ray worklist



1. My work queues: All, Outstanding, Partially endorsed (draft), Fully endorsed, Requested, Urgent.

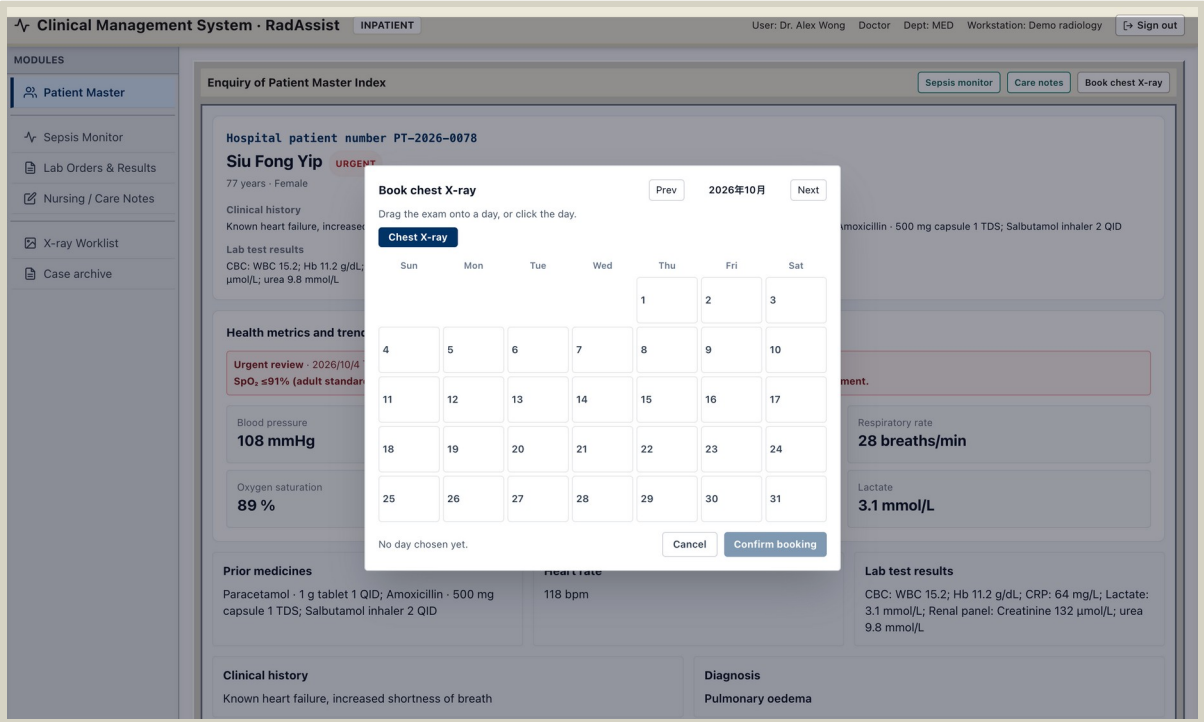
2. Dense RIS-style grid modelled on the COMP4126 lecture worklist. Urgent rows (U) sort first.

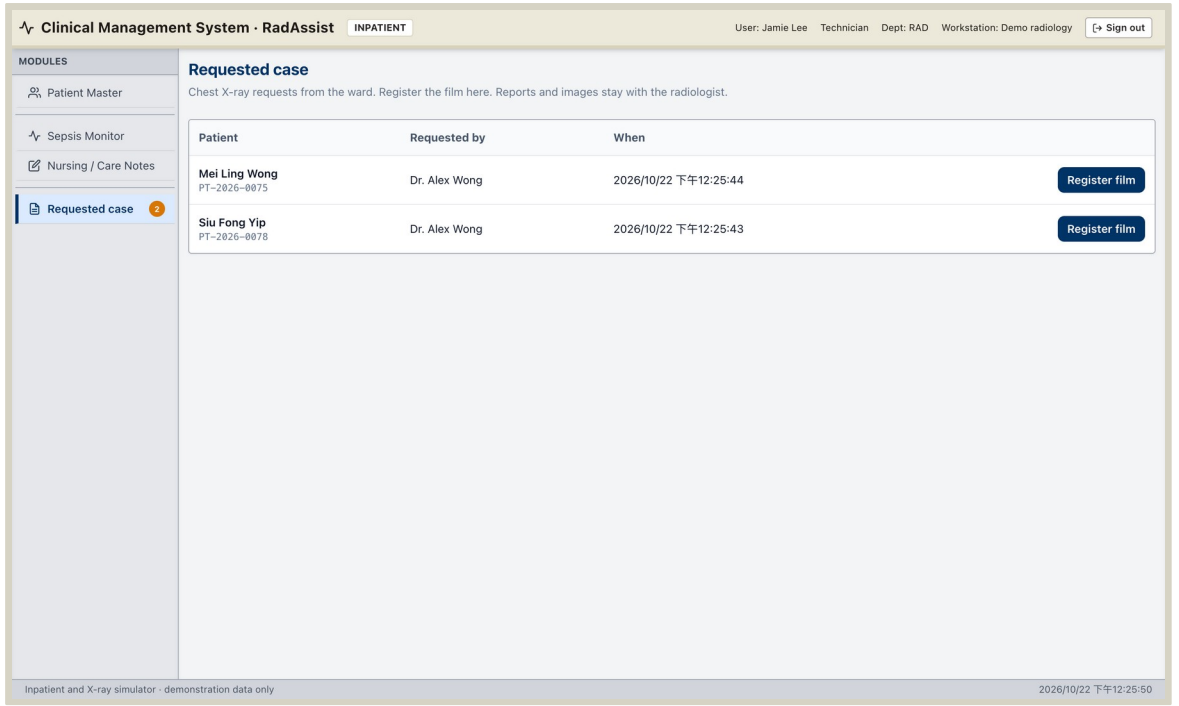
3. Live status: Generating → Pending approve → Endorsed, updated without a reload.

4. Short diagnosis under “Chest”: picked locally from CURV’s text; Azure refines it only after the radiologist edits.

5. Referring doctors use the same grid as “Examination enquiry” — requested and endorsed rows only.

Ordering the exam and registering the film



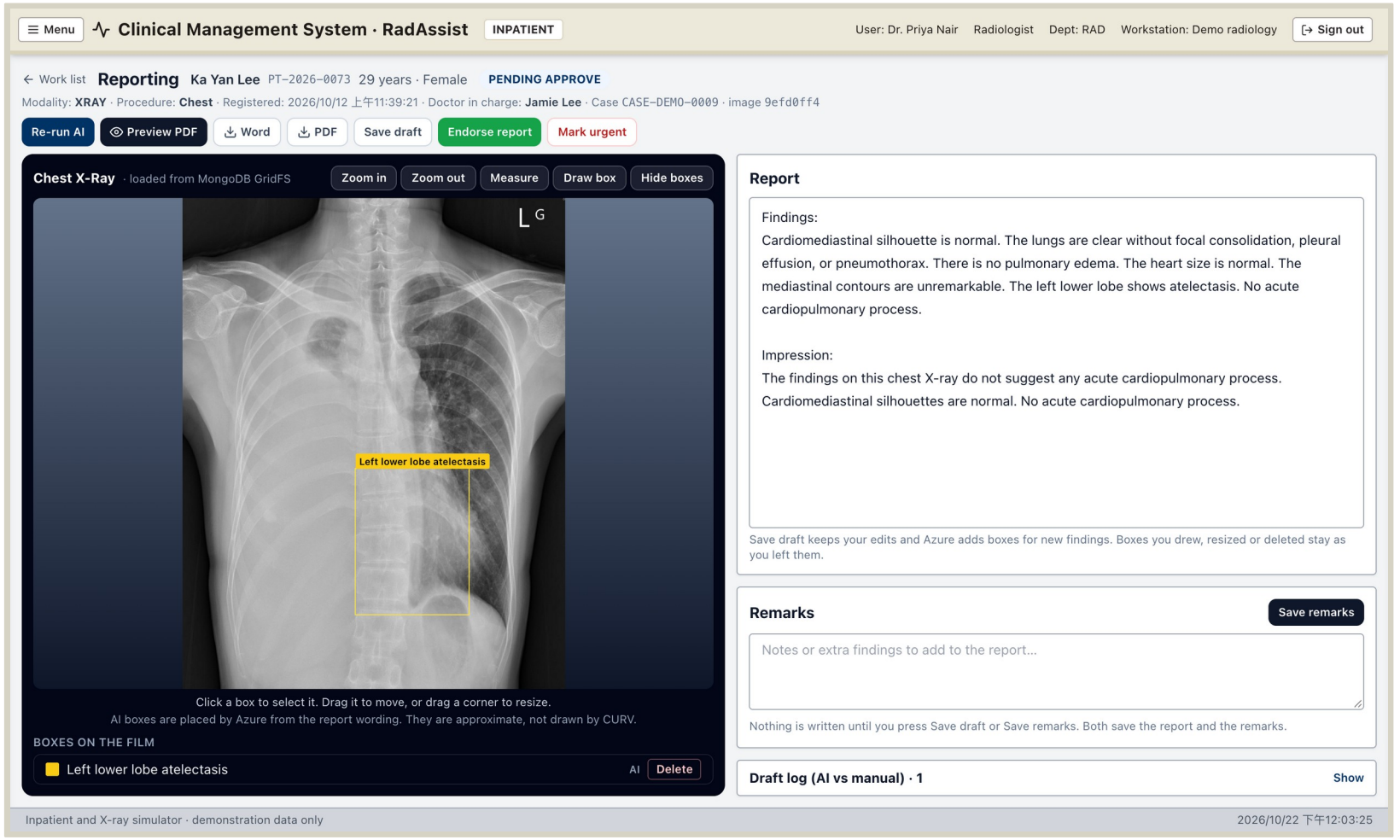


Doctor: drag the Chest X-ray chip onto a day (or click it) → Confirm booking

Technician: Requested case queue → Register film

- One open request per patient; the chart button reads “Requested” until the film is saved.
- DICOM films are rendered to PNG; pixel spacing is kept so measurements can be shown in centimetres.

Report review: film beside an editable draft



1. Toolbar: Re-run AI, Preview PDF, Word, PDF, Save draft, Endorse report, Mark urgent.
2. Film tools: zoom, Measure (cm from DICOM spacing), Draw box, hide boxes / lines.
3. AI boxes placed by Azure from the report wording — select, move, resize, delete or draw your own.
4. Report box is directly editable (Findings + Impression). CURV's hidden Thinking is preserved but never shown.
5. Draft log (AI vs manual) shows exactly which lines, boxes and measurements a human changed.

AI pipeline: CURV writes, Azure assists

CURV — report generation (local)

- Called with the official CURV conversation: system prompt prompt_cxr.txt + “Please analyze this chest X-ray image…”
- Whole frame resized to the model budget (~224 × 224, max_pixels 50176); the review screen keeps the original film.
- Greedy decoding, repetition penalty 1.05, up to 1536 tokens.
- Output tags <findings>, <thinking>, <impression> are saved verbatim — no rewording.
- Thinking is stored with the draft but hidden on every page and in every export.
- If CURV is unavailable the case is still stored and the radiologist writes the report by hand.

Azure OpenAI — assistance (optional)

- **Film boxes** — after the draft and after a wording change; report + film → boxes with severity (normal ones not drawn). Falls back to anatomical zones.
- **Worklist label** — short diagnosis, only after the radiologist has edited the draft.
- **Patient-chart findings** — after endorsement; text-only, Thinking removed, ≤ 6 grouped findings with a severity.
- **Guard-rails** — JSON mode, low temperature, server-side validation; Azure never writes reportText.

Keeping the human in the loop

Draft provenance

Every save appends an editLog row (kind: ai | manual) with added / removed / edited lines, boxes and measurements.

Box merge respects edits

When Azure re-places boxes after a wording change, boxes the radiologist drew, moved, resized or deleted are kept exactly as left.

Honest confidence

Confidence blends report language (hedges vs definite statements) with image support, instead of showing the model's self-reported 100%.

Locking after endorse

Report text, boxes and measurements become read-only; Endorse is refused while AI is running. Remarks and urgency remain editable.

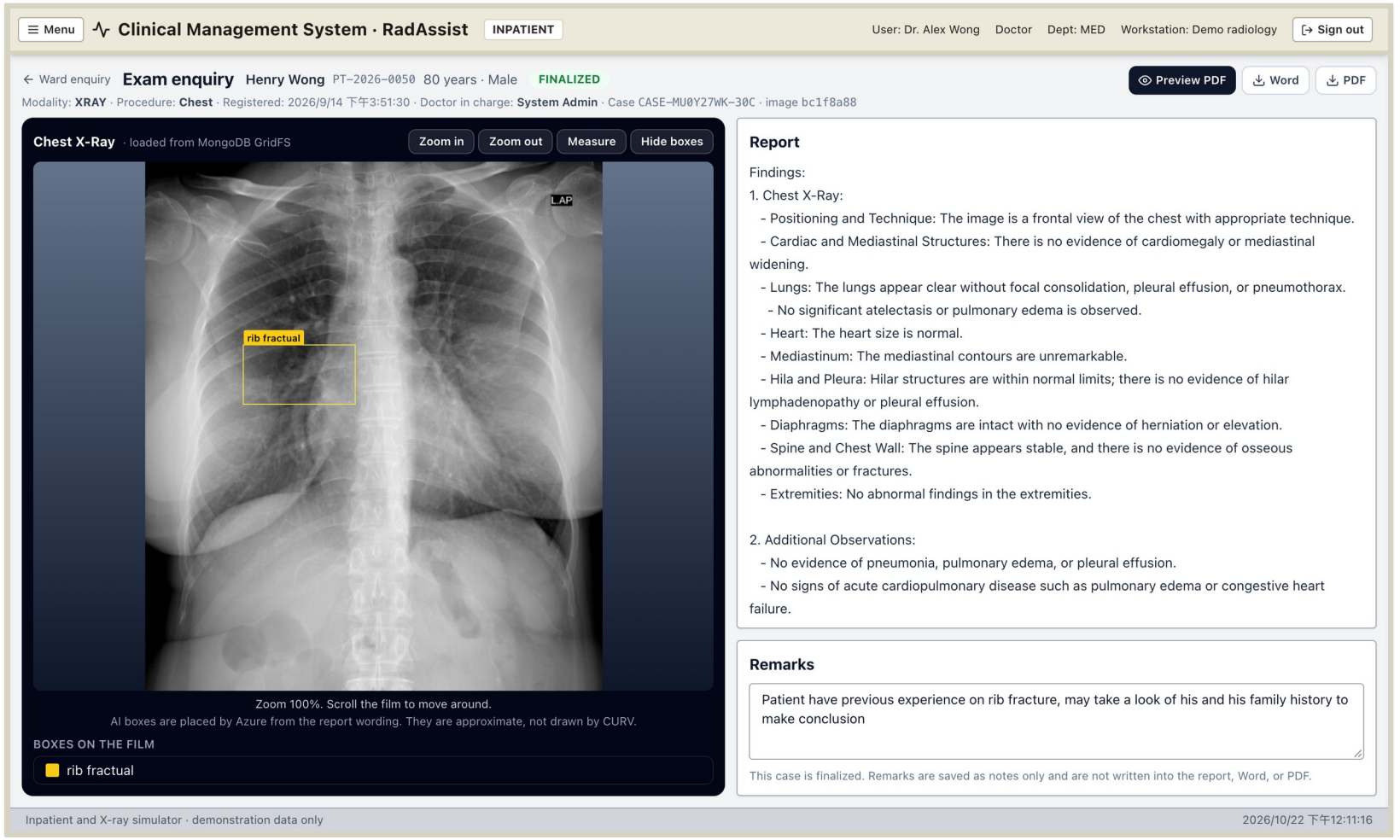
Clean issued report

Doctors, shared links, Word and PDF never receive the draft log, source tags or the Thinking section.

Full audit trail

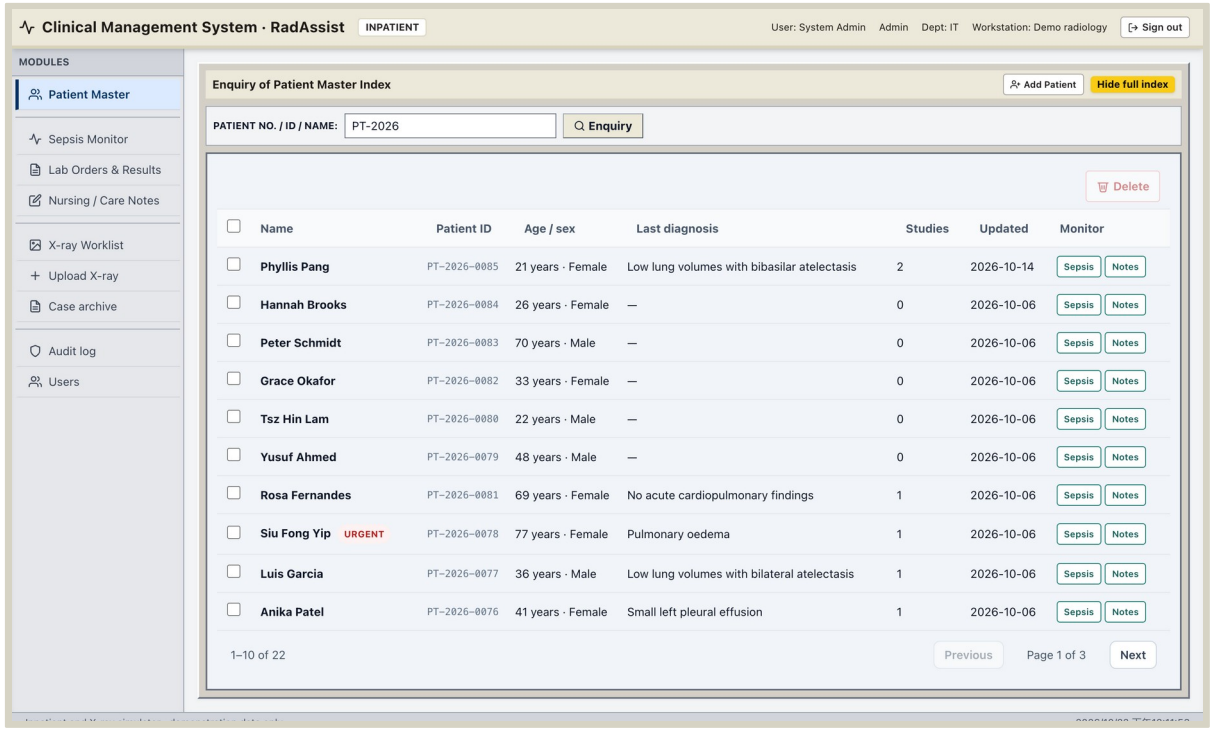
Logins, case and patient changes, AI runs, endorsements, deletions, share links and user changes are logged for the admin.

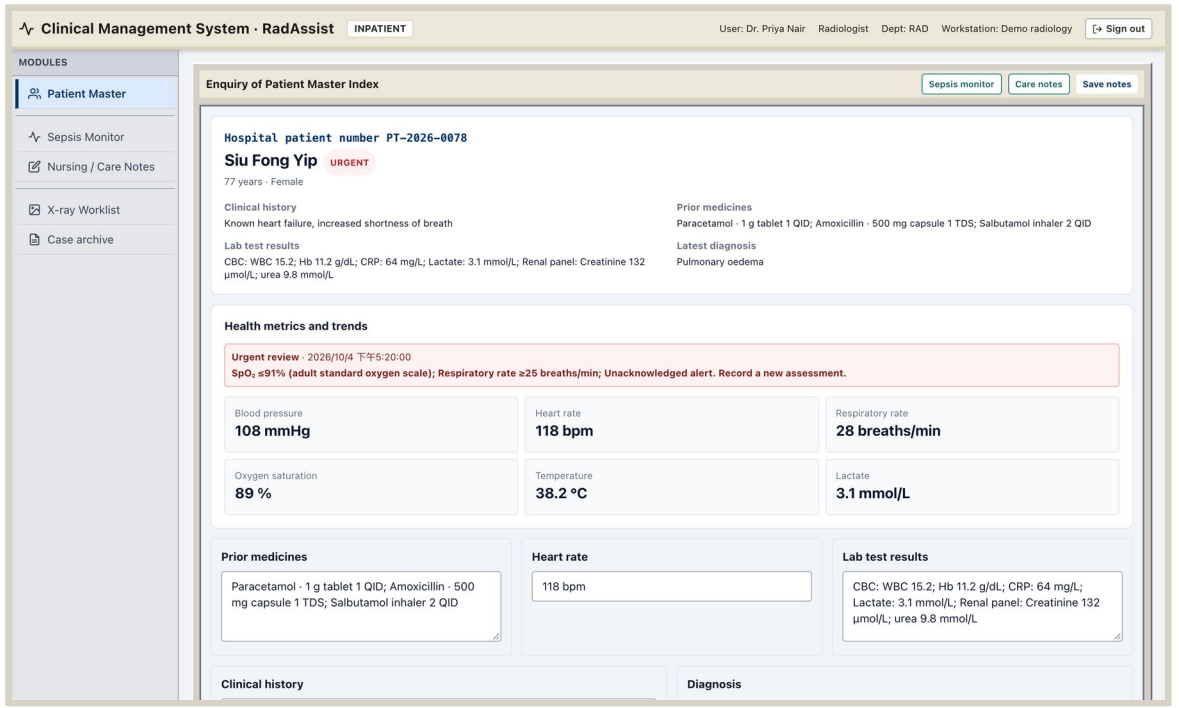
Endorsed report for the referring doctor



- **Read-only** — no re-run, no draft log, no save, no endorse.
- **Clean text** — Findings, Impression, measurements and signature; no Thinking.
- **Film with boxes** — Measure works for reading but is never saved.
- **Export** — Preview PDF, Word (.docx) and PDF include the film, measurement lines, date, reporting doctor and signature.
- **Share link** — a radiologist can create a public token (#/share/<token>) for one endorsed case, and revoke it.

Patient Master Index and patient chart



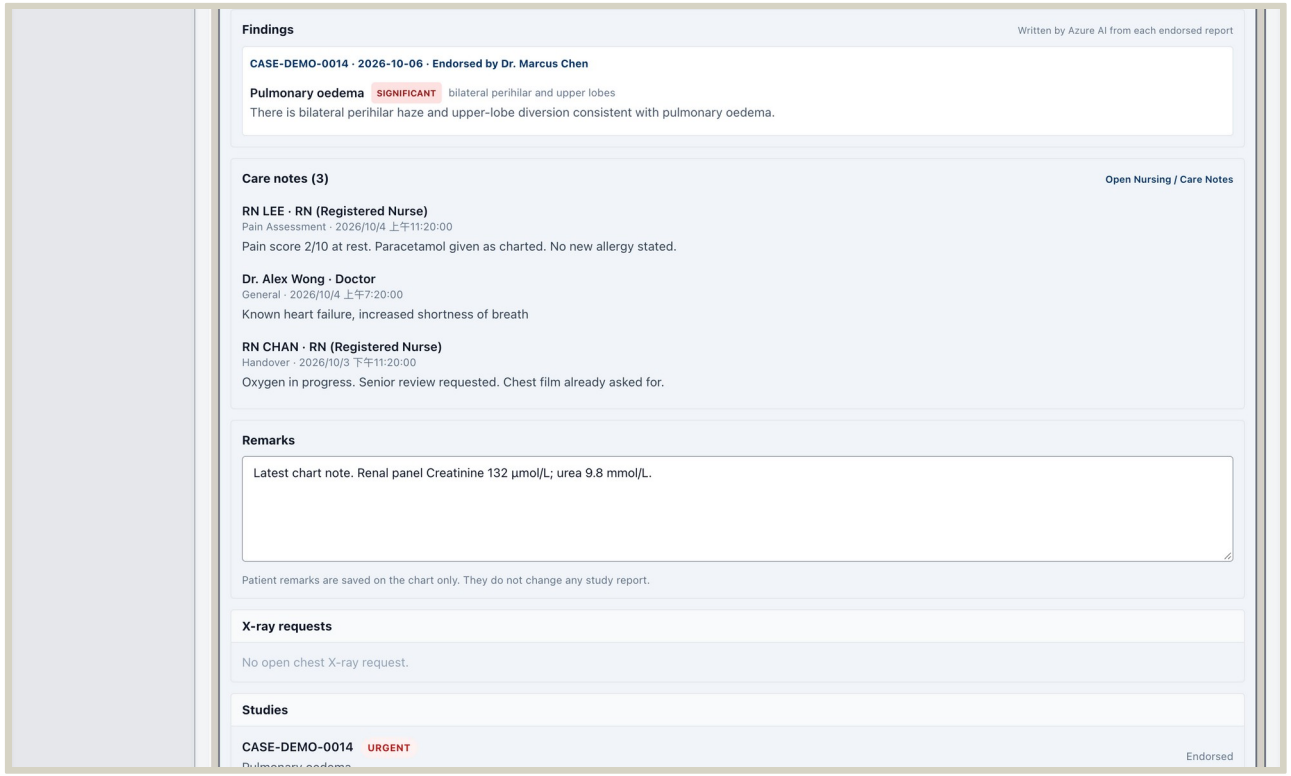


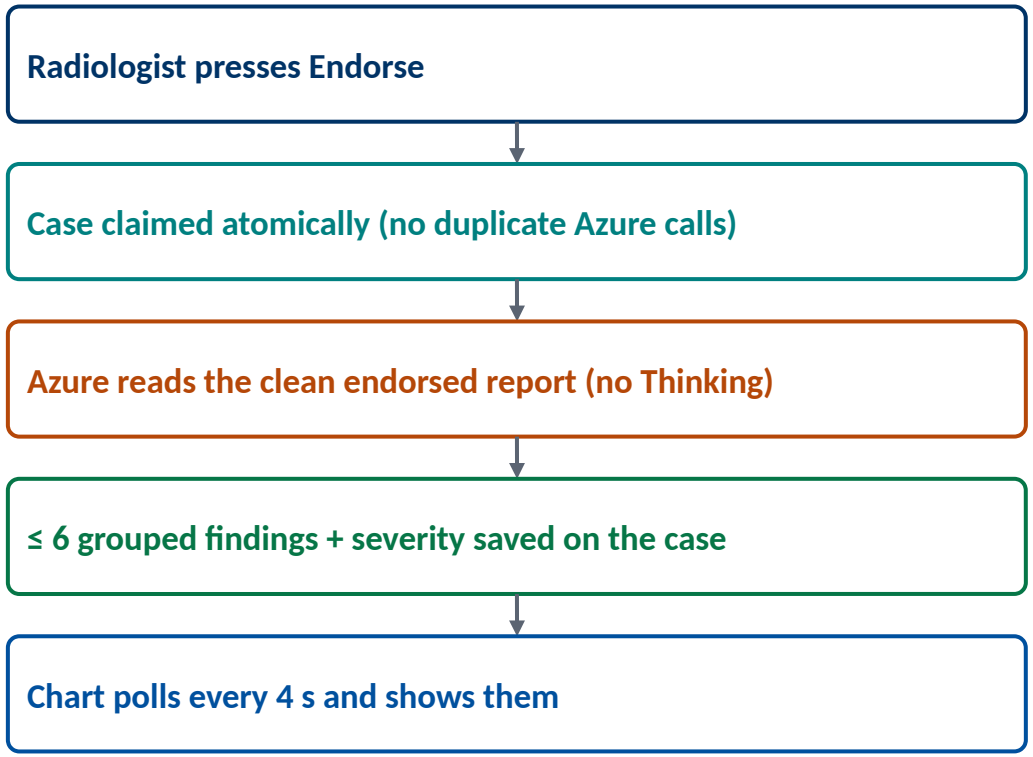
Enquiry of Patient Master Index — search, Sepsis / Notes buttons per row

Chart: history, medicines, labs, latest vitals and review state

- A patient can exist before any X-ray; studies, X-ray requests, diagnosis, remarks and care notes all live on one chart.
- The health-metrics banner flags Urgent / Abnormal review from the latest observations (e.g. $SpO_2 \leq 91$ %, RR ≥ 25).

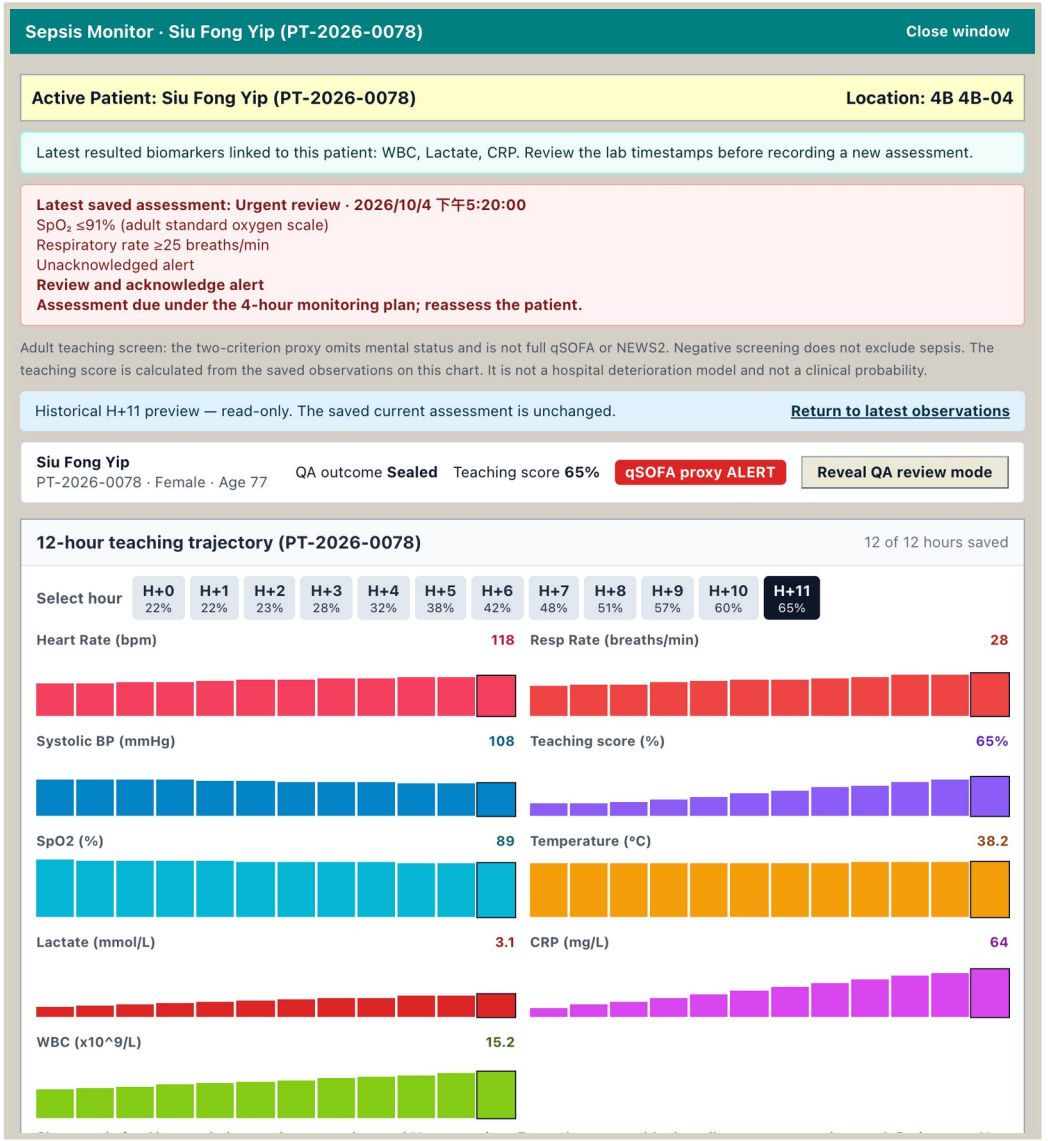
Patient-chart findings written by Azure after endorsement





- Draft reports never appear under Findings — only endorsed studies, each tagged Significant / Minor / Normal with “Endorsed by …”.
- Older endorsed reports are back-filled the first time the chart opens; a lost run (> 3 min) or an Azure failure shows Retry Azure AI to radiologists.
- Hidden from technicians; if Azure is not configured the chart says so instead of guessing.

Inpatient observations & sepsis screening (teaching)



Review states from the CMS sepsis screen

Urgent review: SpO_2 ≤ 91 %, RR ≥ 25, or the two-criterion proxy RR ≥ 22 and systolic BP ≤ 100.

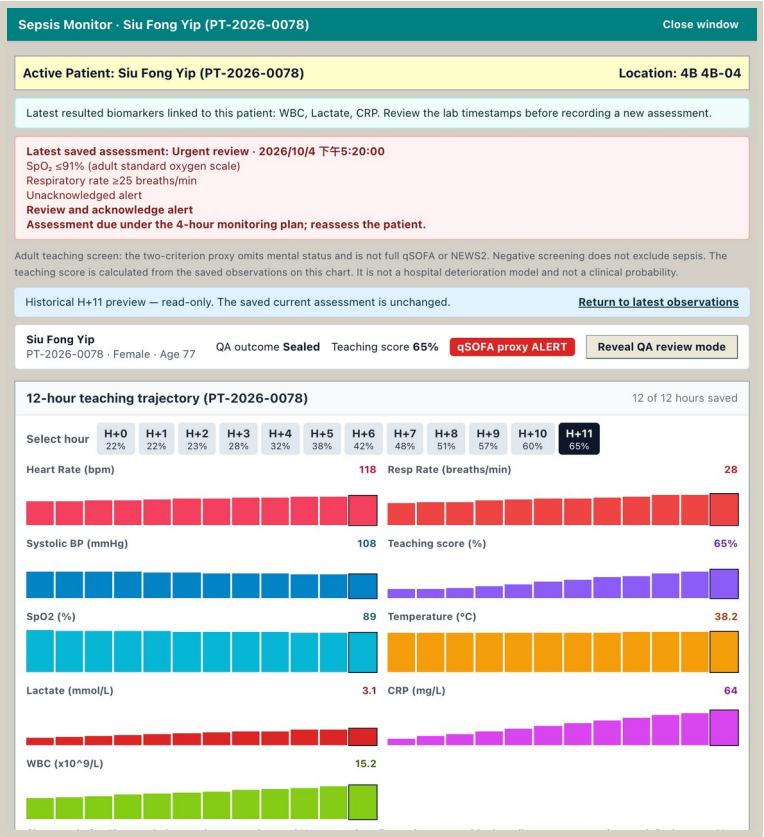
Abnormal: outside adult teaching ranges (HR, RR, BP, SpO_2, temperature, lactate, WBC, CRP).

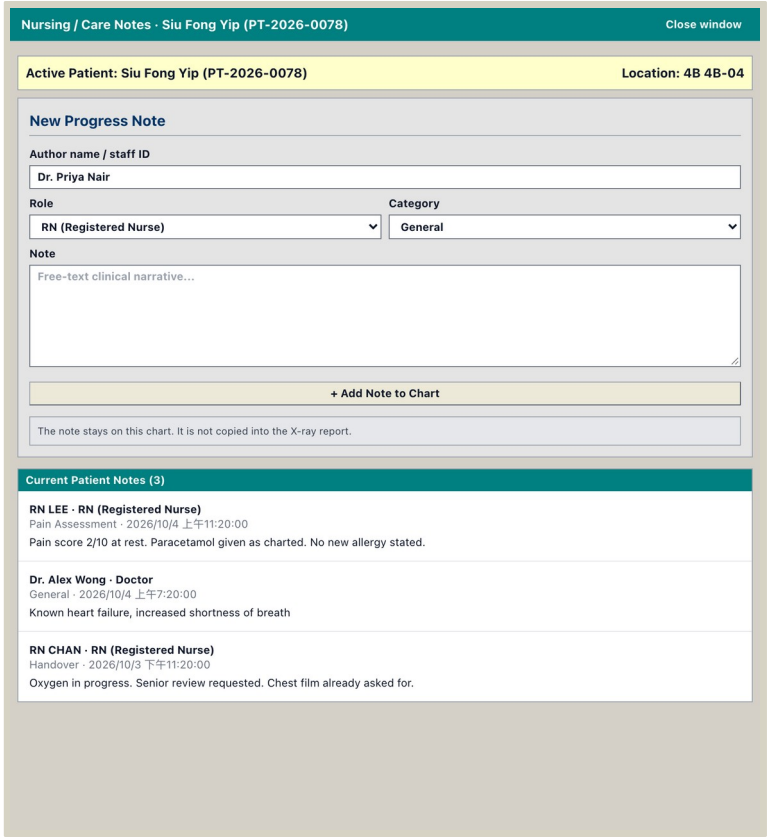
Otherwise: no configured trigger.

- **12-hour trajectory** — per-hour bars for HR, RR, BP, SpO_2, temperature, lactate, CRP, WBC and a teaching score.
- **Hour preview** — step through H+0 … H+11 read-only; the saved assessment is unchanged.
- **Record vitals** — new observations append to the flowsheet and alert audit log.
- **Honest labelling** — the teaching score is not a hospital deterioration model or a clinical probability.

CLINICAL

Sepsis Monitor and Care Notes in their own windows

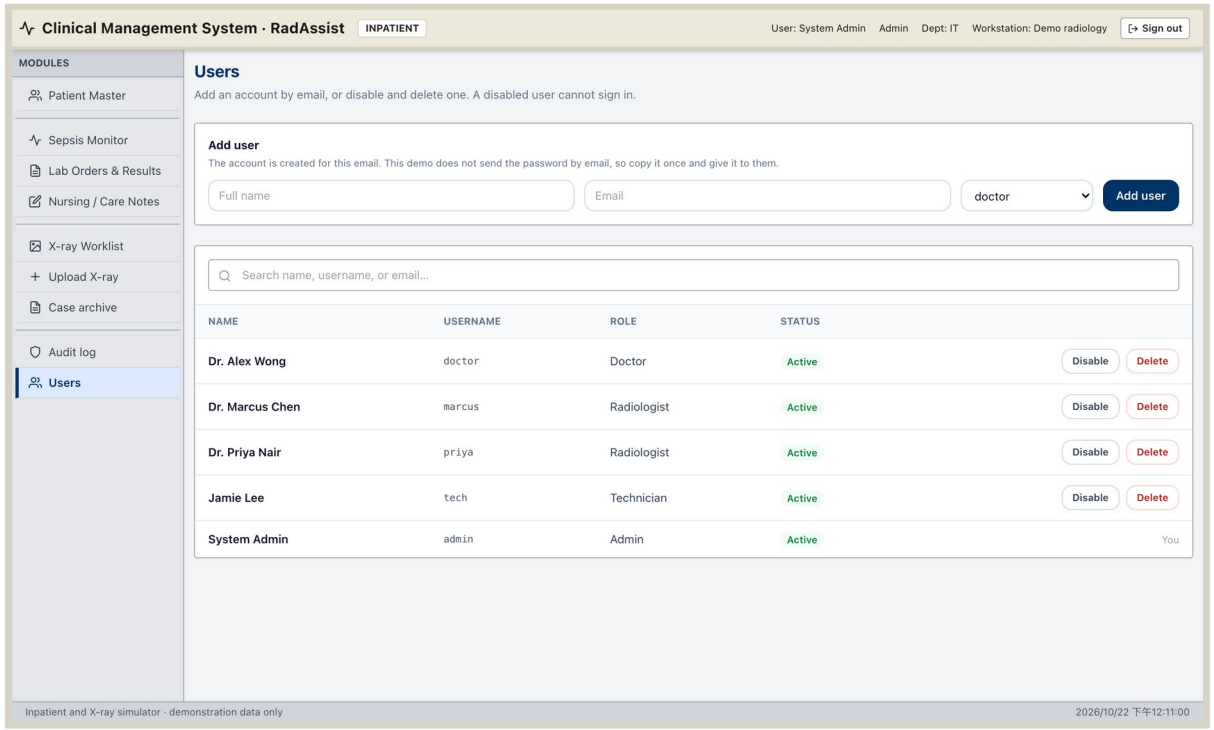


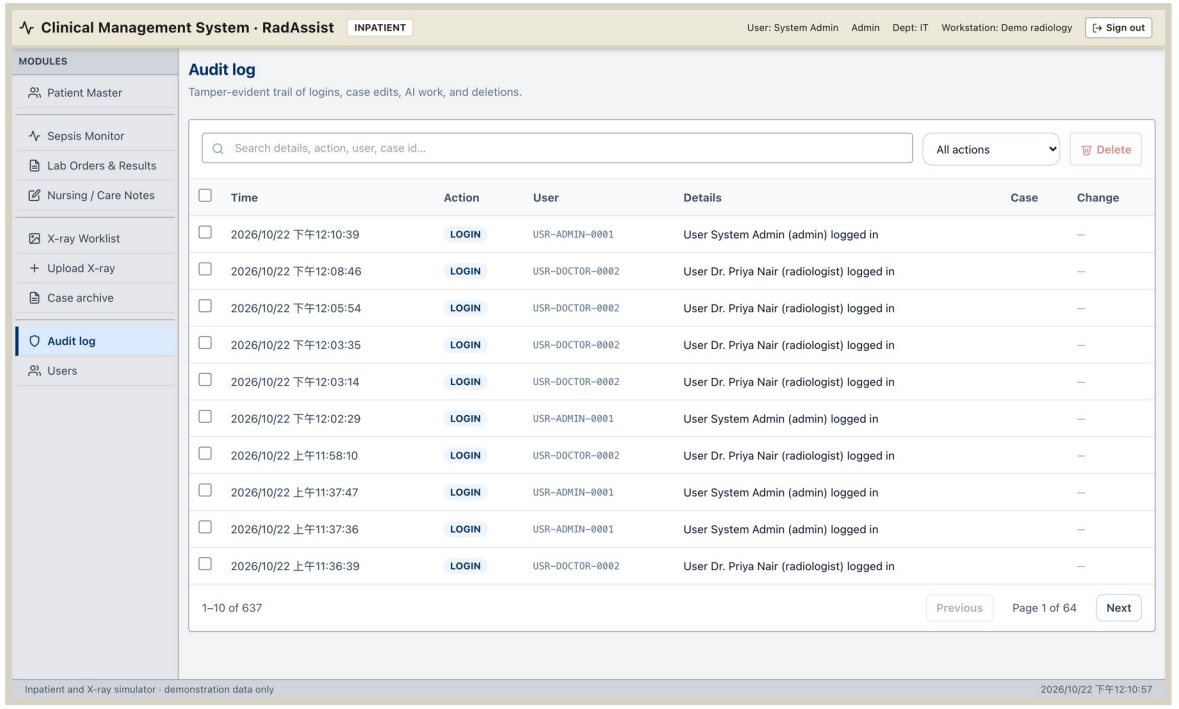


- **Read side by side** — Sepsis monitor and Care notes open the patient in a separate browser window — beside the chart or on a second screen.
- **One patient only** — no ward worklist or patient picker, so nothing is recorded against the wrong patient.
- **Live sync** — vitals or notes saved in a window are posted back (same-origin postMessage) and the chart updates without a reload.
- **Window reuse** — each view has its own named window; opening another patient reuses it.
- **Pop-up blocked?** — the same view opens as a draggable, resizable floating panel instead.

`?view=sepsis&patientId=…` · `?view=notes&patientId=…`

Administration: users and audit trail





Users: add by email, one-time temporary password, enable / disable / delete

Audit log: filter by action, user, case; admin-only

- Safeguards: an admin cannot disable or delete their own account or remove the last active administrator.
- Bulk delete (cases, patients, audit rows) removes related studies and GridFS images; a patient delete keeps nothing orphaned.

Data model

Patient

patientId · first / middle / last name
age · sex · history · remarks
medicines · heartRate · labResults
ward · bed · admissionStatus
observations[] · labOrders[] · careNotes[]

1 : N

Case (one study / film)

caseId · patientId · status · urgent
reportText · remarks · diagnosis(+Source)
findings[] = film boxes · measurements[]
analysisState · boxesState · editLog[]
chartFindings[] · chartFindingsState
imageId (GridFS) · pixelSpacingMm
requestedBy · createdBy · finalizedBy/At · shareToken

User

username · email · name
role: technician | radiologist | doctor | admin
bcrypt password · isActive

AuditLog

userId · action · details
affectedCaseId · before / after
timestamp

Case status	requested	pending	pending_approve	finalized
Shown as	Requested	Outstanding / Generating	Pending approve (draft)	Endorsed (locked)

Images are streamed from the GridFS bucket "images" and require a valid JWT (except a share link's single film).

Security and privacy

In place

- JWT authentication; every route checks the role on the server.
- Response stripping per role (technicians never receive film, report or findings; doctors never receive draft logs or source tags).
- bcrypt password hashing; temporary passwords shown once.
- Helmet security headers; GridFS images require a JWT.
- Share links are random 24-byte tokens for one endorsed case, revocable.
- Chart windows accept updates only from the same origin.
- Secrets in backend/.env (never committed); no HKID or address collected.

Known gaps (demo scope)

- CORS is intentionally loose for local Vite ports.
- Public demo sign-up (POST /api/auth/register) should be disabled for real use.
- JWT is stored in localStorage; httpOnly cookies would resist XSS better.
- Seed passwords are public in the repository and must be rotated on shared databases.
- Patient ID allocation (max + 1) can collide under concurrent creates.
- Synthetic chart data only — MIMIC-IV vitals and labs are not loaded.

EVALUATION

Testing and verification

193	19	35	~14.7k
automated UI tests	test suites, all passing	REST route handlers	lines of JavaScript

- **UI behaviour tests** — linkedom DOM + mocked API: every review-page control (edit, boxes, measurements, endorse, share, Word / PDF), role restrictions, worklist, pagination, DICOM, audit, users, share page.
- **Patient chart tests** — Azure findings only for endorsed reports, running / failed / Retry states, hidden from technicians, Care notes, separate windows, sync back to the chart, blocked pop-up fallback.
- **Backend logic checks** — atomic claim of chart findings against a throw-away MongoDB: two simultaneous loads → one Azure call; lost runs restart; Retry forces a re-run.
- **End-to-end** — test-e2e-case.mjs logs in against the running API and checks every case, its report text, findings and image.
- **Live checks** — Azure prompts verified on sample reports (Thinking removed, grouped findings); screens verified in headless Chrome per role.

Limitations

Not a clinical device

Decision support only — not a PACS, not CE / FDA cleared, not a substitute for a radiologist.

Model input resolution

CURV sees a ~224 px film; fine detail is lost. The original is kept for human review.

Approximate boxes

Azure boxes come from the report wording and the film, not from CURV's attention; they can be imprecise.

Measurement accuracy

Centimetres depend on DICOM pixel spacing; other films measure in pixels.

Synthetic chart data

Vitals, labs, medicines and the sepsis score are teaching data, not MIMIC-IV.

Single-machine AI

CURV runs on one Apple Silicon Mac — the critical dependency for any live demo.

Future work (Semester 2 and beyond)

- **Hardening** — tighten CORS, remove public sign-up, httpOnly cookie sessions, rotate secrets, structured logs with request IDs.
- **Backend test suite** — API black-box tests for Express + MongoDB, plus upload-to-draft latency measurements.
- **Deployment** — static frontend + Node API in a container, Cosmos secrets in a key vault, CURV on a dedicated Apple Silicon / GPU host, Azure via private endpoint.
- **Richer imaging** — full DICOM / PACS viewer with compressed transfer syntaxes and windowing.
- **Bidirectional binding** — hover a sentence to highlight its box and vice versa; global confidence threshold slider.
- **Evaluation with clinicians** — time-to-report and edit-distance study comparing AI-drafted vs blank-page reports; usability feedback from radiologists.
- **Real clinical data** — link MIMIC-IV vitals and labs under an ethics plan to replace synthetic chart data.

DEMO

Live demo plan (≈ 10 minutes)

1. **Doctor** Book a chest X-ray on the calendar
2. **Technician** Requested case → register the film
3. **Radiologist** Worklist: Generating → Pending approve
4. **Radiologist** Review: Azure boxes, edit text, resize/draw a box, measure
5. **Radiologist** Save draft → show draft log → download PDF
6. **Radiologist** Endorse → open chart → Azure writes Findings
7. **Any role** Open Sepsis monitor + Care notes windows; record a vital and a note
8. **Doctor** Read the clean endorsed report; export Word / PDF
9. **Admin** Users, audit trail, share link

Before the session: start backend, Vite and start_server.sh; check /api/health (mongo up) and :8001/health; wait for the model to load.

Thank you

Questions & discussion

RadAssist AI
AI drafts. The radiologist decides. The ward sees only what was signed.

PANG Ho Yiu · 23226609 · HKBU Computer Science